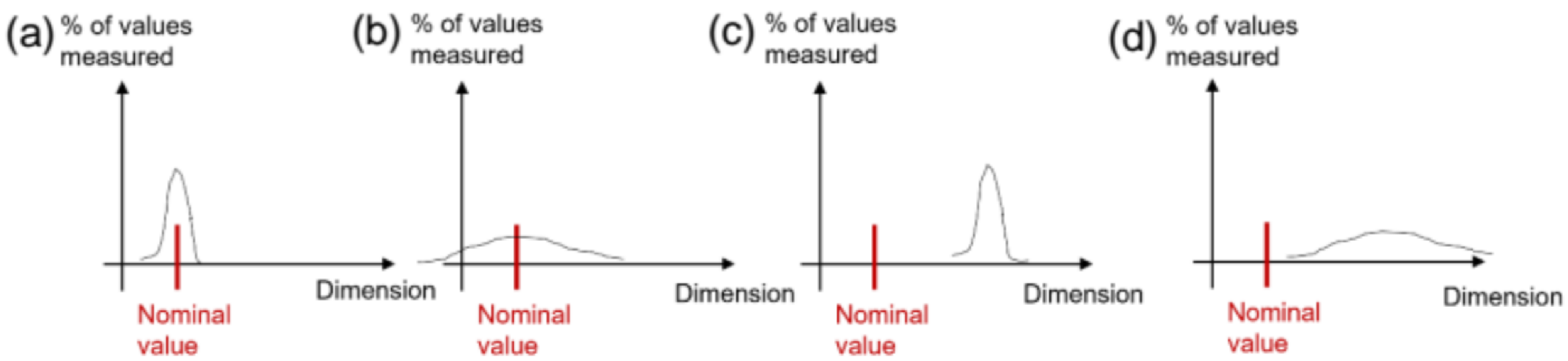
Q1: The following diagrams represent the % of value measured as a function of the dimension of a fabricated part. Which diagram illustrates that the measuring method was precise but not accurate?



c

Q2: A stylus apparatus measured the following deviations at the surface of a flat metal piece. Calculate the surface roughness of that metal piece.

Deviation in μm
+50
-45
+80
+65
-33
-85
+78
+95
-250
+65

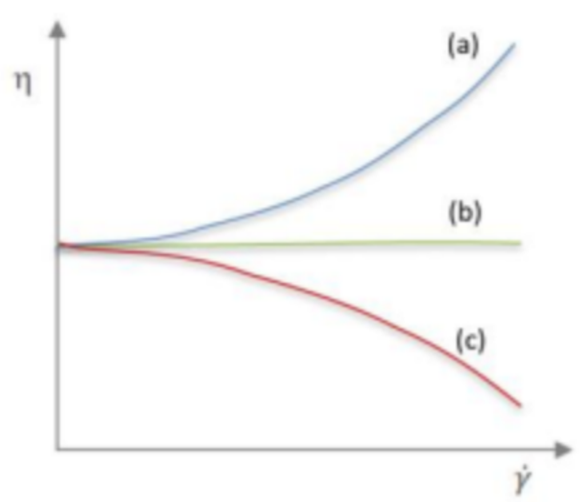
$n = 10$

$\sum x = 50 - 45 + 80 + 65 - 33 - 85 + 78 + 95 - 250 + 65$

$= 20$

$\bar{x} = \frac{20}{10} = 2\,\mu m$

Q3: The following graph shows the viscosity as a function of the shear rate for 3 fluids. Which curve represents a shear-thinning fluid?



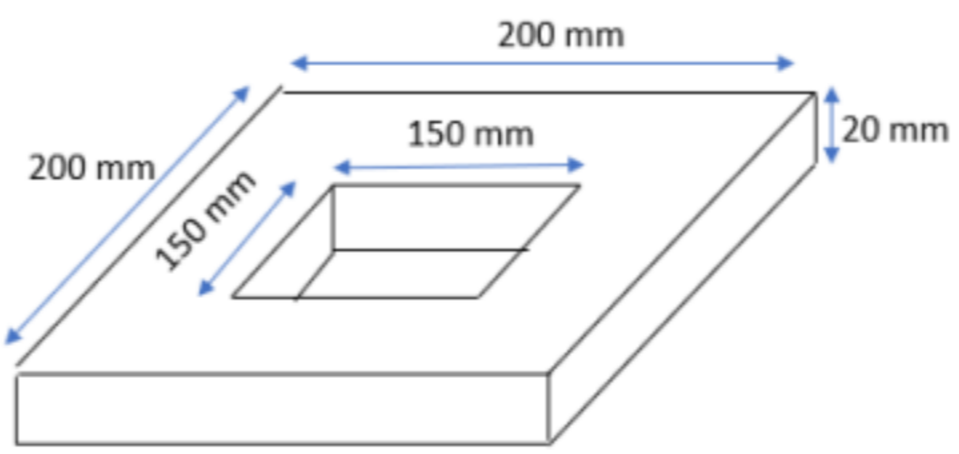
c

Q4: What is the advantage of a three plate mould over a two plate mould?

(a) It is cheaper.
(b) It separates the part from the sprue and the runner.
(c) It has a cooling system.

b

Q5: The following piece is to be fabricated using sand casting. Knowing the mould constant is 0,4 min/mm^2, calculate the total solidification time in minutes.



$$T_{ST} = C_m \left(\frac{V}{A}\right)^2$$

$$V = 200^2 \times 20 - 150^2 \times 20$$
$$= 350000 \, mm^3$$

$$A = 2 \times (200^2 - 150^2)$$
$$+ 200 \times 20 \times 4$$
$$+ 150 \times 20 \times 4$$
$$= 63000 \, mm^2$$

$$T_{ST} = 0.4 \left(\frac{350000}{63000}\right) = \frac{1000}{81}$$
$$\approx 12.34 \, min$$

Q6: What happens during the cooling down of a casted part made of iron with high C content?

(a) Formation of cracks due to thermal contraction and shrinkage
(b) Formation of particulates due to graphitization
(c) Dewetting of the solidified metal from the mould leaving a gap between the cast part and the mould
(d) Formation of an inner cavity.

b

Q7: What is the purpose of internal chills?

(a) To make holes inside the final part.
(b) To facilitate the demolding.
(c) To control the solidification rate and prevent tearing.
(d) To change the metal flow from turbulent to laminar.

c

Q8: A compound die will be used to blank and punch a large washer out of 6061ST aluminum alloy sheet stock 4.00 mm thick. The outside diameter of the washer is 55.0 mm and the inside diameter is 16.0 mm. Determine the punch die size for the punching operation. The tensile strength of the material = 310 MPa and its corresponding metal allowance, Ac = 0.07.

Ans (mm): 16.0

Q9: Which of the statements is **ODD** one out? (this means the statement that is correct, if the 3 other statements are wrong, or the statement that is wrong if the 3 other statements are correct).

(a) The final flow stress is commonly used to estimate or compute the rolling force.
(b) In flat rolling, the draft can be determined from the final thickness.
(c) The springback in bending process is due to the release of remained elastic energy in the bent part. ✓
(d) V-bending is generally inexpensive and generally used for high production. ✗

d

Q10: The starting flat blank sheet has a size of 70 mm in length and 50 mm in width. The blank sheet has a 4mm thickness. The sheet is bent to the profile having a bend radius of 6mm as shown in Figure 1. The profile lengths, L1 and L2, are 33 and 30 mm respectively. Determine θ (in degrees).

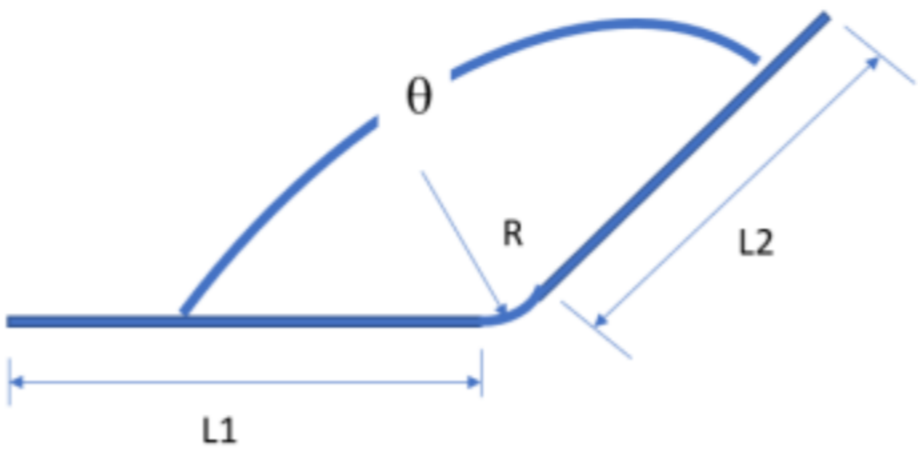


Figure 1: Bent Profile

Ans: 113.2°

Remaining length $= 70 - 33 - 30$
$= 7\text{mm}$

$s = r\alpha$

$\alpha = \frac{s}{r}$

$= \frac{7}{6}$

$\theta + \alpha = 180^\circ$

$\theta = 180 - \frac{7}{6} \times \frac{180}{\pi}$

$= 113.1549239^\circ$

$\approx 113.2^\circ$